1Breadcrumb 2025 C# Fullstack Technical Hiring Assessment

Introduction

Welcome to the 1Breadcrumb technical test.

This portion of the technical test is take-home. Spend up to an hour on the task, but no more. If you have only partially completed the exercise by the end of an hour, that is ok and you'll have an opportunity to talk through and extend your work in the subsequent interview stages, including a pair-programming exercise. The intention of this exercise is to sample your practical coding skills.

Your task is to build a small web application using C# & React based on a low fidelity design sketch, an incomplete API Spec & some contextual background.

Once complete, please share a copy of your work with us either as a git repo or as an archive. In a subsequent interview stage we will be doing a pair programming exercise to extend your project together. You will not be penalised at all for having AI generated code, but do ensure to organise, structure and understand your work so you are ready to answer questions about it and pair on it with other humans.

Definitions

- **Crumb(s)**: A 1Breadcrumb (1BC) employee or team member (think your favourite loaf of bread, or a delicious, perfectly balanced sour-dough).
- **Daily Cafe**: A daily, optional, company wide 15min event for the purpose of simulating 'grabbing a coffee with your workmates' in a hybrid work environment. Many people will grab themselves a coffee to drink while attending for a bit of chit-chat on whichever topics come up (e.g. social meetups, work ideas, weekend plans, dog training).

The Exercise

You were attending Daily Cafe a couple of days ago and an idea came up to start a company-wide Crumb-to-Crumb book lending library (the Library). You thought this idea sounded great and you've been informally talking about it with your team ever since. One of the
challenges that you and your team identified to making this idea real, was a book discovery problem. No one knows who has which books, and which ones are available for lending!

Today you received a Slack message from the designer in your team sharing a sketch (see below concept sketch) of a fairly simple web-app which they think could solve the discovery problem! They've asked if you can spend an hour of time today to build a prototype which you can then demo at Daily Cafe tomorrow to collect feedback on before taking it any further.

Rules

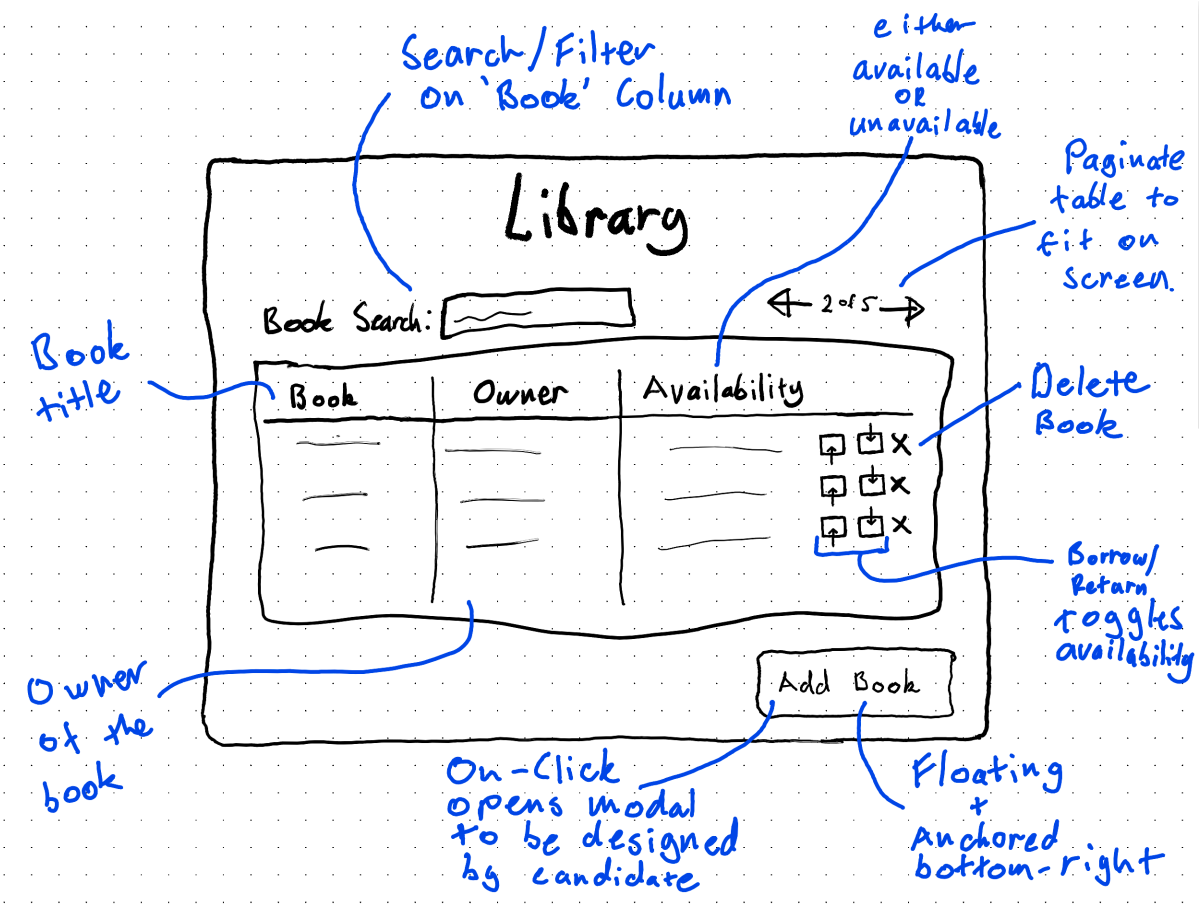
1. Spend *1 hour maximum* on the task.
2. You *must* use C# & React.
3. You *can* use any additional coding languages, frameworks and libraries you would like.
4. You *can* use AI - but please keep in mind this task will become the basis for future interview rounds so you need to be able to answer questions about your code, and be prepared to extend your codebase in pair-programming with other humans where you won't have the use of AI.

Hints

- Reach out to 1Breadcrumb if you have any questions at any time.
- Try to break the page into well-sized, maintainable components.
- Make sure your code is of a quality where you'd be comfortable giving it to your fellow developers for review & pair programming in real life.
- Prioritise coverage of functionality over aesthetics. Note: usability is NOT considered aesthetic.

- Feel free to tweak the design to make it easier to build or if you have any design ideas you'd like to try out as long as the core use-case can still be achieved. If you do this, please be ready to explain what and why you made changes when you come to the interview.
- For any complex logic and/or any critical components, write some automated tests *OR* explain to us in comments how you plan to test those pieces of code *OR* why you chose not to write any tests.

Concept Sketch



Incomplete API Spec

- https://libapi.1breadcrumb.com/